

ECO387C - Fall 2024

Problem Set 10

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1 - Lucas Tree

Consider the setup of the Lucas tree model from class. Specifically, suppose that there is a durable asset that pays dividends d_t that follows a Markov chain with transition probabilities given by $\pi(d'|d)$. At the beginning of each period t , the representative household's holdings of this asset are denoted by a_t and they choose a_{t+1} , or how many units of this asset to hold for next period. The price of this asset is denoted by p_t , which is taken as given. I.e., the representative consumer maximizes:

$$\mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t \frac{c^{1-\sigma} - 1}{1-\sigma} \right\}$$

subject to the budget constraint

$$c_t + p_t a_{t+1} = (d_t + p_t) a_t$$

1. Let $v(a, d)$ denote the representative consumer's value function. Write the Bellman equation for the representative consumer's problem.
2. Define this economy's recursive competitive equilibrium.
3. Let T denote the Bellman operator on the RHS of the Bellman equation you found in Part 1. Prove that T satisfies Blackwell's sufficiency conditions for a contraction mapping.
4. Let $p(d)$ denote the price of this asset when the current state is given by d . Show that in equilibrium, $p(d)$ solves the following equation:

$$p(d) = \beta \sum_{d'} \left(\frac{d'}{d} \right)^{-\sigma} [p(d') + d'] \pi(d'|d) \quad (1)$$

5. Suppose that dividends are now I.I.D. over time. Solve the functional equation Equation 1 for $p(y)$. (**Hint:** Guess that $\frac{p(d)}{d^\sigma} = \alpha$, where α is a constant.)
6. Suppose that now, $u(c) = \ln(c)$, $\beta = 0.96$, and that d_t follows a 2-state Markov chain. Specifically, $d_t \in \{d_L, d_H\} = \{0.9, 1.1\}$, and $P(d' = d_L | d = d_L) = p = 0.4$, and $P(d' = d_H | d = d_H) = q = 0.3$. Solve for $p(d_L)$ and $p(d_H)$.